Generative Modeling for Speaker Diarization

ADASP Formal Presentation

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Outline

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Overview

What is Speaker Diarization?

Objective: Decide who speaks when

- Who: *Identify* the active speakers.
- When: Find the *boundary* of their speech/voice activity.

Output: Rich Transcription Time Marked (RTTM) files

A csv file where each line contains an audio segment’ s information:
file type, file name, channel ID, begin time, duration, speaker ID.

```
SPEAKER ES2005a 1 0.000000 7.800000 <NA> <NA> 20 <NA> <NA>
SPEAKER ES2005a 1 7.800000 5.760000 <NA> <NA> 13 <NA> <NA>
SPEAKER ES2005a 1 13.560000 6.720000 <NA> <NA> 20 <NA> <NA>
```

Overview

Particularity of SD:

- ~~supervised learning / classification problem: finite number and in-domain speakers.~~
- Unsupervised learning / clustering problem: indetermined number, out-of-domain speakers

Comparison to other audio tasks:

- voice activity detection (binary classification, indifferent to speaker identity)
- automatic speech recognition (closed-class, supervised learning, soft annotations)
- sound event detection (more general than verbal activities)
- acoustic unit discovery (acoustic unit instead of speaker identity)

Evaluation

Diarization Error Rate (DER)

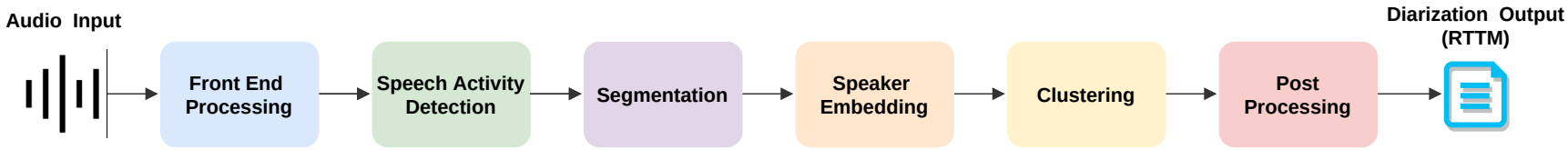
$$\mathrm{DER} = \frac{\text{false alarm} + \text{missed detection} + \text{speaker confusion}}{\text{ground truth duration}}$$

- False Alarm: a frame is annotated to speech while it isn’ t.
- Missed: a frame is not annotated to speech while it is.
- Speaker Confusion: a frame is annotated to one speaker while it is another speaker.

Specialties

- Who: identifiers are pseudo rather than absolute labels.
  - transform predicted IDs to target IDs that maximizing the overlap.
- When: boundaries are not consistent across different annotators.
  - add forgiving collar to both ends of segments (usually 0.25s).

General Pipeline

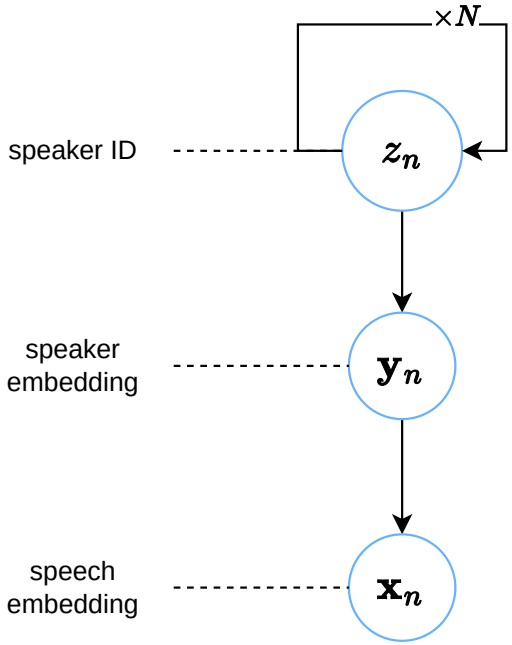


- Traditional System: modular optimization but correlated
- Jointly optimization system: merge several blocks with joint optimization objectives
- End-to-end system: deep neural network

Particularly we focus on VBx system.

The generative model for SD in VBx

Graph model



Formulation

$$
\begin{aligned}
p(\mathbf{X}, \mathbf{Z}, \mathbf{Y}) &= p(\mathbf{X}|\mathbf{Z}, \mathbf{Y})p(\mathbf{Z})p(\mathbf{Y}) \\
&= \prod_n p\left(\mathbf{x}_t|z_t\right) \prod_n p\left(z_t|z_{t-1}\right) \prod_s p\left(\mathbf{y}_s\right)
\end{aligned}
$$

Interpretation

At each time frame t, a speaker $z_t = s$ is activated according to a **conversation model** $p(z_t|z_{t-1})$, whose speaker embedding $\mathbf{y}_s$ lies in a **speaker space**, then a noisy observation of speaker embedding, i.e. the speech segment embedding $\mathbf{x}_t$ is emitted from a **speaker model** $p(\mathbf{x}_t|z_t = s) = p(\mathbf{x}_t|\mathbf{y}_s)$.

Hence the speaker diarization task is equivalent to solve $p(\mathbf{Z}|\mathbf{X})$

Speaker space: feature extraction

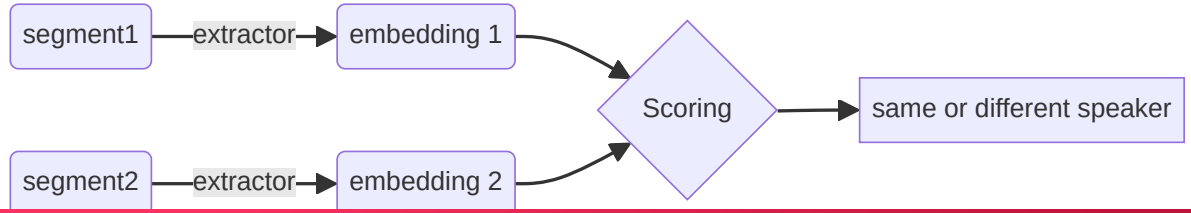
Representation and comparison of audio segments

Speech is a mixture of different information: acoustic, semantic, prosody, **speaker identity**, etc.

Embedding

- i-vector: traditional representation extracted by factor analysis
- x-vector: robust neural representation trained on large augmented corpus (noises, reverberation…)
  - Frontend: MFCC, filter banks, Wav2Vec, WavLM…
  - Backend: ResNet, ECAPA-TDNN…
  - Loss: cross entropy, aam loss, circle loss…

Scoring



Speaker Model: PLDA

How speakers are distributed in the feature space

Formulation

We need to deal with two types of variability of the representation: different speakers and different speech segments of the same speaker.

$$P(\mathbf{y}) = \mathcal{N}(\mathbf{y} \mid \mathbf{m}, \Phi_b)$$
$$P(\mathbf{x} \mid \mathbf{y}) = \mathcal{N}(\mathbf{x} \mid \mathbf{y}, \Phi_w)$$

Φ_b: the between-class variance, characterize how speaker differs from each other.
Φ_w: the within-class variance, characterize how speech differs within the same speaker.

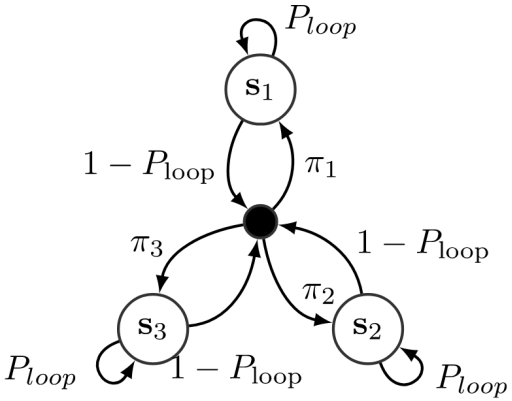
About naming

- Linear discriminant analysis by Gaussian Mixture point of view
- Generalize mixture prior from a finite sum of Dirac to a continuous Gaussian

Conversation model: Markov Chain

How speakers take turns in the conversation

Topology



Formulation

$$p(z_t = s|z_{t-1} = s') = p(s|s') = (1 - P_{loop})\pi_s + \delta(s = s')P_{loop}$$

- Homogeneous along the time
- one speaker per time frame, unable to deal with multiple speakers (overlap speech)

VBx = HMM + PLDA

Combined Model

$$
\begin{aligned}
p(\mathbf{X}, \mathbf{Z}, \mathbf{Y}) &= p(\mathbf{X}|\mathbf{Z}, \mathbf{Y})p(\mathbf{Z})p(\mathbf{Y}) \\
&= \prod_t p\left(\mathbf{x}_t|z_t\right) \prod_t p\left(z_t|z_{t-1}\right) \prod_s p\left(\mathbf{y}_s\right)
\end{aligned}
$$

where

$$
\begin{aligned}
p(\mathbf{y}_s) &= \mathcal{N}(\mathbf{y}_s; \mathbf{0}, \mathbf{I}) \\
p(\mathbf{x}_t|z_t = s) = p(\mathbf{x}_t|\mathbf{y}_s) &= \mathcal{N}(\mathbf{x}_t; \mathbf{V}\mathbf{y}_s, \mathbf{I}) \\
p(z_t = s|z_{t-1} = s') &= p(s|s') \\
p(s|s') &= (1 - P_{loop})\pi_s + \delta(s = s')P_{loop}
\end{aligned}
$$

Variational Inference

In order to compute the intractable $p(\mathbf{Z}|\mathbf{X}) = \int p(\mathbf{Z}, \mathbf{Y}|\mathbf{X})d\mathbf{Y}$, we need variational inference.

1. Surrogate model $q(\mathbf{Y}, \mathbf{Z})$ for $p(\mathbf{Z}, \mathbf{Y}|\mathbf{X})$, compute the ELBO function

$$\mathcal{L}\left(q(\mathbf{Y}, \mathbf{Z})\right) = E_{q(\mathbf{Y},\mathbf{Z})}\left\{\ln\left(\frac{p(\mathbf{X}, \mathbf{Y}, \mathbf{Z})}{q(\mathbf{Y}, \mathbf{Z})}\right)\right\}$$

2. Mean-field approximation to factorize $\mathbf{Y}, \mathbf{Z}$, i.e. $q(\mathbf{Y}, \mathbf{Z}) = q(\mathbf{Y})q(\mathbf{Z})$
3. Together with the independency $\mathbf{Y} \perp\!\!\!\perp \mathbf{Z}$, we can further factorize it to

$$\mathcal{L}\left(q(\mathbf{Y}, \mathbf{Z})\right) = F_A E_{q(\mathbf{Y},\mathbf{Z})}\left[\ln p(\mathbf{X}|\mathbf{Y}, \mathbf{Z})\right] + F_B E_{q(\mathbf{Y})}\left[\ln\frac{p(\mathbf{Y})}{q(\mathbf{Y})}\right] + E_{q(\mathbf{Z})}\left[\ln\frac{p(\mathbf{Z})}{q(\mathbf{Z})}\right]$$

with two hyper-parameters as a counter-effect of our assumptions.

Variational Inference

4. Solving the functional calculus equation $\frac{\delta \mathcal{L}}{\delta q} = 0$, we get

$$\ln q(\mathbf{Y}) = \frac{F_A}{F_B} E_{q(\mathbf{Z})} \left[\ln p(\mathbf{X}|\mathbf{Y}, \mathbf{Z}) \right] + \ln p(\mathbf{Y}) + const$$
$$\ln q(\mathbf{Z}) = F_A E_{q(\mathbf{Y})} \left[\ln p(\mathbf{X}|\mathbf{Y}, \mathbf{Z}) \right] + \ln p(\mathbf{Z}) + const$$

where the surrogate prior $q(Y)$ and $q(Z)$ are mutually dependent, hence can be iteratively optimized.

5. (VBEM) Iteratively optimize $q(\mathbf{Y})$ and $q(\mathbf{Z})$:

- crucial quantity: $q(z_t = s)$, marginal distribution of speaker at time t.
- $q(\mathbf{Y})$ part: relatively easy, product of Gaussian factors $q(\mathbf{Y_s})$.
- $q(\mathbf{Z})$ part: a surrogate HMM hence involving Baum-Welch algorithm.
- Fact: GMM-HMM is solvable by EM, PLDA-HMM as a generalization is solvable by VBEM

6. When ELBO converges, $\arg\max_s q(z_t = s)$ is the speaker identity alignment.

Future work

VBx plays as a baseline

1. Playing with speaker model: regularization, disentanglement
2. Playing with conversation model: deep sequential models instead of HMM
3. Do everything in end-to-end way (EEND): large synthetic data, discriminative training
4. More signal processing inductive bias:
   - α-stable process
   - multi-channel spatial information
5. Hint from other level of information: semantic, emotional, articulatory.

Meanwhile, I try to avoid 💩⛰️ codebases (Kaldi, speechbrain)…

END